Software Requirement Specification (SRS)

Requirements Feasibility Report

Project Name: Campus Pulse– Student Activities & Career Hub Application
Date: 2026-10-04
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1. Introduction

- **Purpose:** Evaluate the technical, operational, and structural feasibility of building "Campus Pulse," a unified web application for publishing student club announcements, managing campus events, and displaying targeted notices from both the **Career Centre** and **Student Activities Office**.
- **Scope**:
  - Multi-role publishing dashboards for Career Center, Student Activities, and Club Leaders.
  - Moderation and approval workflow for club-submitted events.
  - Student event discovery feed with filtering, RSVPs, and calendar syncing.
  - QR code digital ticketing and organizer check-in system.
  - Targeted push notifications with customizable subscription channels

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2. Overall Description

2.1 Product Functions

The core capabilities of the system are grouped by operational workflow:

- **Role-Based Publishing & Moderation:**
  - Post announcements and event listings tailored to specific organizational entities.
  - Moderate and approve, request edits for, or reject pending student club event submissions.
  - Broadcast priority alerts across official campus channels.
- **Event Discovery & Attendance Lifecycle:**
  - Search, filter, and bookmark events based on categories, host bodies, and dates.
  - Manage event RSVPs and export bookings to external calendars.
  - Generate dynamic student QR passes and validate them via offline-resilient scanning tools.
- **Notification & Preferences Management:**

- Filter push notification feeds based on explicit user-selected interests and channels.

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3. Specific Requirements

3.1 Functional Requirements

Career Center & Administrative Features

- **FR-CC-01 Corporate Postings:** The Career Center shall create event postings tagged by industry sector, job type (internship/full-time), and major applicability.
- **FR-SA-01 Global Notices:** Student Activities administrators shall issue system-wide announcements with priority delivery flags.
- **FR-SA-02 Moderation Queue:** The system shall hold club-submitted events in an unpublished queue until approved by Student Activities.

Club Leader Features

- **FR-CL-01 Event Submission:** Club Leaders shall submit event proposals including title, date, time, location, max capacity, and poster attachments.
- **FR-CL-02 Check-in Scanner:** Club Leaders shall scan attendee QR codes using their mobile device camera to track attendance.

Student Features

- **FR-ST-01 Discovery Feed:** Students shall view a consolidated feed of approved club events, Student Activities notices, and Career Centre opportunities.
- **FR-ST-02 Filtering:** Students shall filter events by host (Career Centre / Student Activities / Clubs), date, and category tags.
- **FR-ST-03 Ticket Generation:** Upon RSVP, the system shall generate a unique, dynamic QR code pass stored in the user's app profile.

3.2 Non-Functional Requirements

- **NFR-PERF-01 Latency:** The system shall render the event feed in under $200\ \text{ms}$ under a baseline load of 1,000 active concurrent users.
- **NFR-SEC-01 Access Control:** The system shall enforce strict role-based authorization ensuring only authorized administrators can approve events or publish global notices.
- **NFR-AVAIL-01 Uptime:** The core API service shall maintain 99.9% operational availability during active academic terms.

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3. System Context & Constraints Assessment

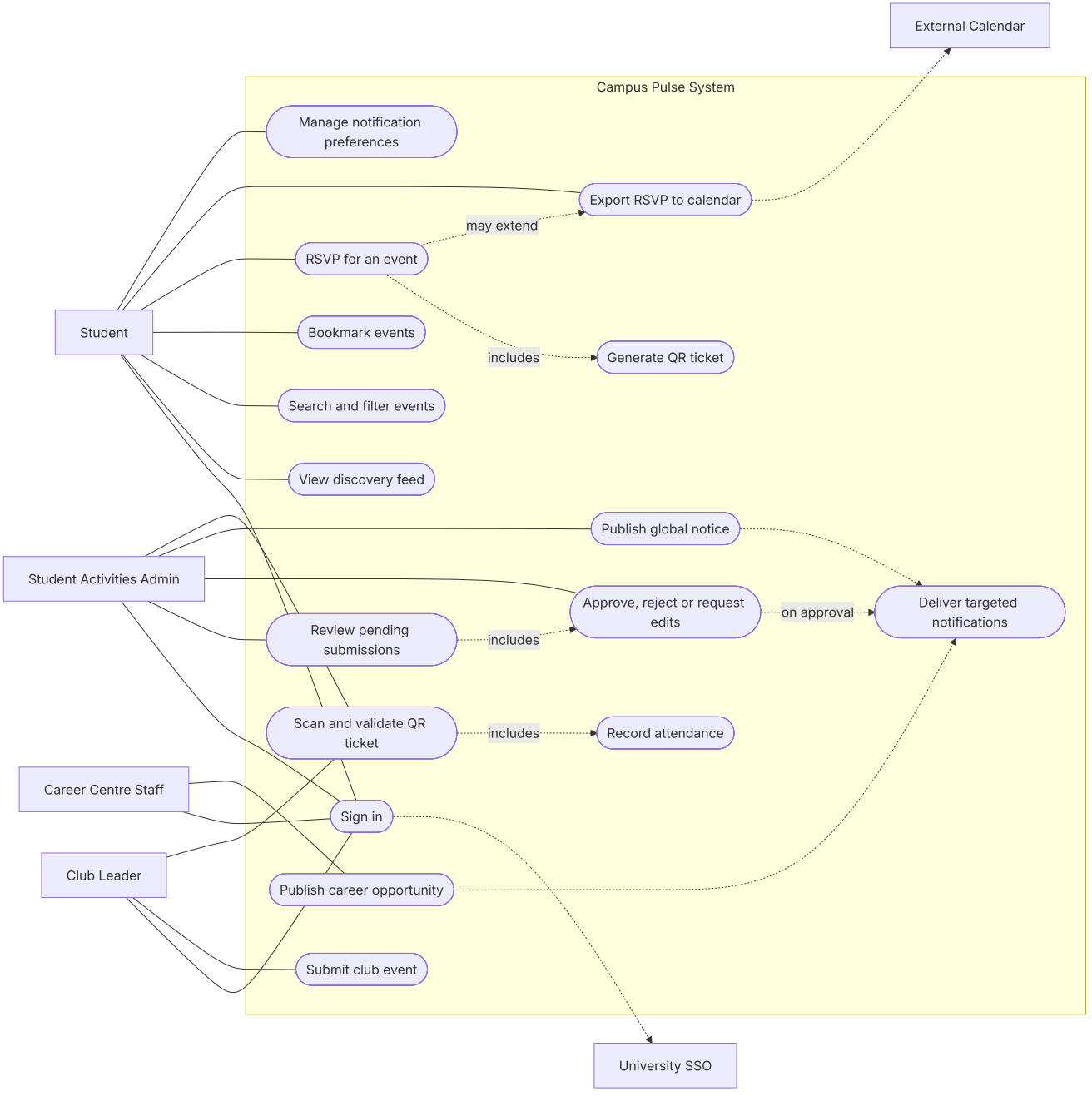
Interface / Component	Proposed Requirement	Technical Viability	Risk Level (Low/Med/High)	Mitigation / Notes
University SSO	Single sign-in via central campus authentication	Pass	Low	Standard OAuth2 / SAML endpoint integration.
Calendar Sync	Export event RSVP directly to device/Google Calendar	Pass	Low	Google/Outlook OAuth.
Push Notifications	Instant alerts for urgent Career & Event notices	Pass	Medium	Implement rate-limiting / daily digests to prevent spam.
Media Attachments	High-res PDF/image attachments for posters & CVs	Pass	Medium	Use cloud storage bucket with strict file size caps (max 10MB).

Req ID	Proposed SRS Requirement Description	Verifiable?	Achievable?	Priority	Feasibility Status
FR-01	**Role-Based Postings:** Career Center, Student Activities, and Clubs have separate dashboard views and posting privileges.	Y	Y	Essential	**Feasible**
FR-02	**Event Approval Workflow:** Club-created events require automated or manual approval from Student Activities before appearing publicly.	Y	Y	Essential	**Feasible**
FR-03	**Career Center Category Filters:** Filter events by Internships, Job Fairs, Resume Workshops, and Industry Sectors.	Y	Y	Essential	**Feasible**
FR-04	**QR Code Event Check-in:** Student Activities and Clubs can scan student QR codes at venue doors to verify attendance.	Y	Y	Essential	**Feasible**
FR-05	**AI Event Recommendations:** Recommend events based on student interest tags, major, and past attendance history.	Y	N	Optional	**Unfeasible (Phase 2)**
FR-06	**In-App Messaging:** Direct peer-to-peer chat between students and club leads regarding event questions.	Y	Conditional	Conditional	**Deferred (Risk of Abuse)**

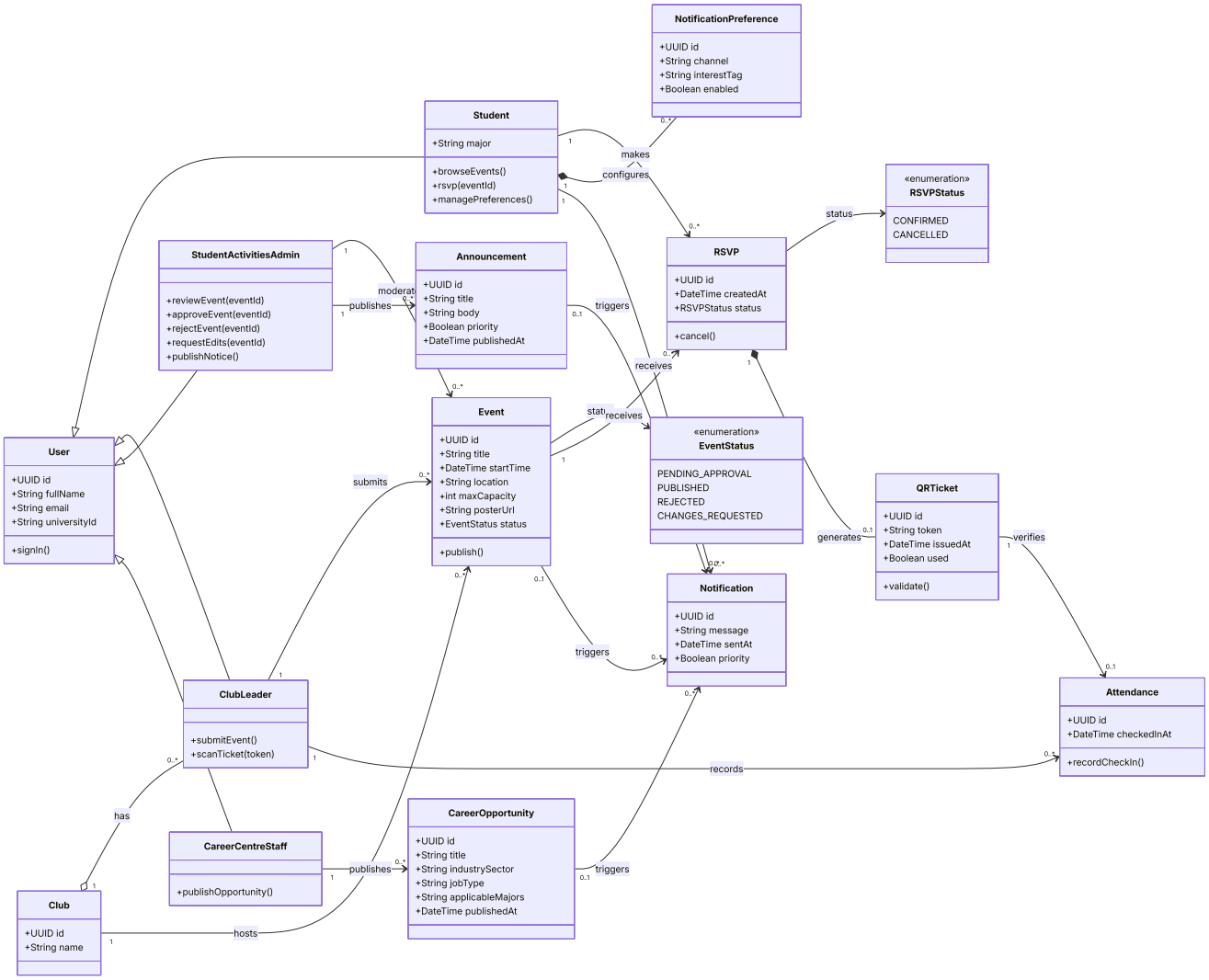
Non-Functional & Performance Feasibility

Requirement	Target Metric	Proposed Architecture	Verdict
Peak Concurrency	Support 2,000 active concurrent users during campus-wide announcements	Containerized Microservices + Caching Layer (Redis)	**Feasible**
Notification Latency	Push alerts delivered within 5 seconds of publishing urgent Career Center notices	Firebase Cloud Messaging (FCM) / APNs	**Feasible**
Data Privacy (GDPR/Local)	Attendance data and CV submissions restricted to authorized staff/leads only	Role-based data sanitization & encrypted S3 storage	**Feasible**

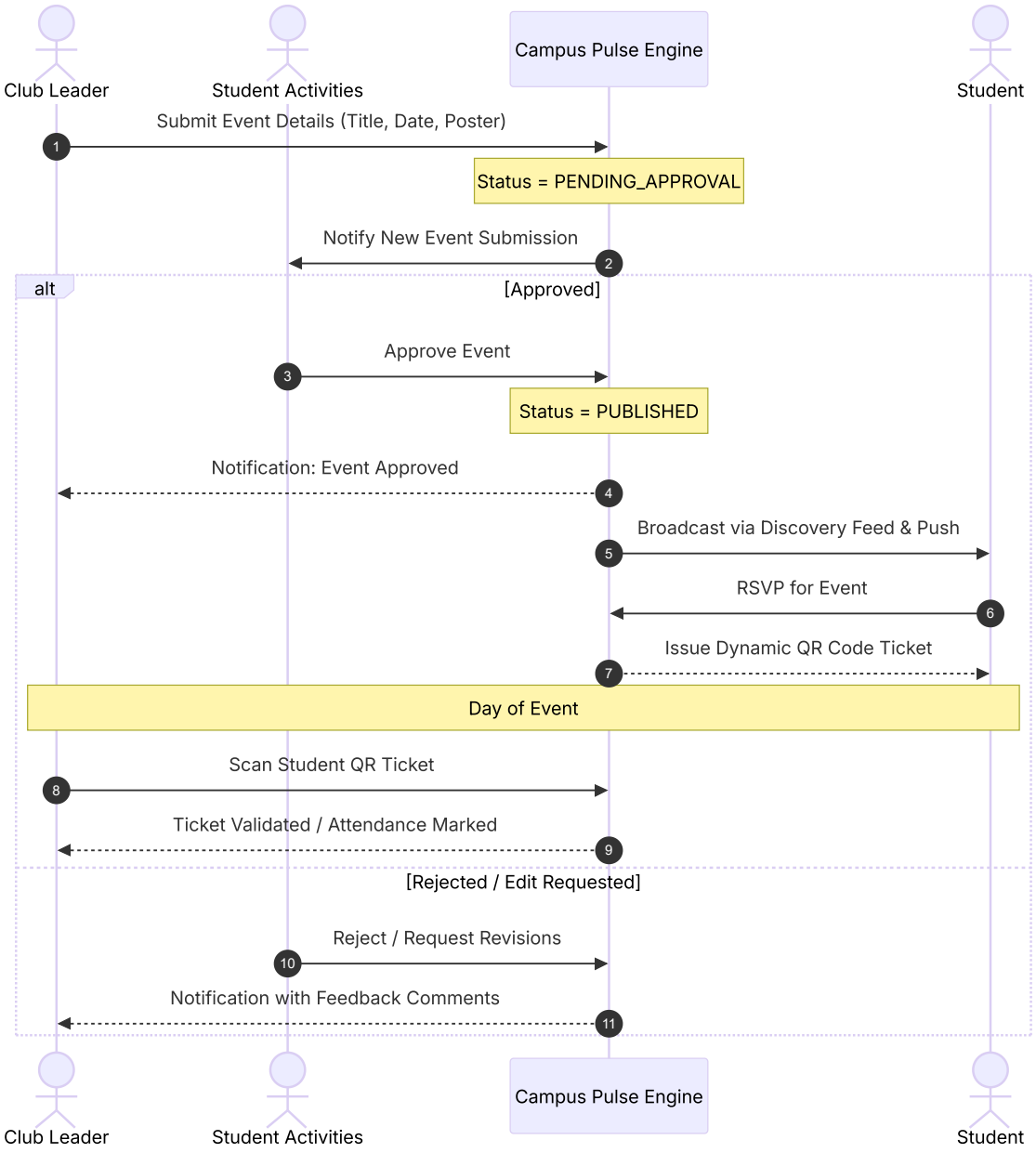
4. Use Case Diagram



5. Class Diagram



6. Sequence Diagram


Club Leader
Student Activities
Campus Pulse Engine
Student
Submit Event Details (Title, Date, Poster)
1
Status = PENDING_APPROVAL
Notify New Event Submission
2
alt
[Approved]
Approve Event
3
Status = PUBLISHED
Notification: Event Approved
4
Broadcast via Discovery Feed & Push
5
RSVP for Event
6
Issue Dynamic QR Code Ticket
7
Day of Event
Scan Student QR Ticket
8
Ticket Validated / Attendance Marked
9
[Rejected / Edit Requested]
Reject / Request Revisions
10
Notification with Feedback Comments
11
Club Leader
Student Activities
Campus Pulse Engine
Student